

FUN WITH ELECTRICITY!

Voltage Divider & Potentiometer: Manual Dimmer Fun

1. WATER PIPE ANALOGY & CONCEPTS

Imagine turning a water faucet or tap: turning it lets more water flow out, and turning it back slows it to a trickle! Electricity works in a very similar way inside circuits.

Component	What It Does
Voltage Divider	Splits up or divides electricity in a path so we can control how much energy goes to a light bulb or component.
Potentiometer	A special Magic Knob ! Turning it changes resistance and controls how easily electricity flows.
Resistors in Series	Two or more resistors connected end-to-end to help control electrical current step-by-step.

The Dimmer Effect:

- **Turn Clockwise:** Allows MORE electricity to flow → Light gets **BRIGHTER!**
- **Turn Counter-Clockwise:** Allows LESS electricity to flow → Light gets **DIMMER!**

2. HANDS-ON EXPERIMENT: BUILD THE VOLTAGE DIVIDER CIRCUIT

Materials Needed:

- Breadboard
- 10kΩ Potentiometer (Knob)
- LED Light (or 3V Bulb)
- 330Ω Fixed Resistor
- Jumper Wires
- 9V Battery & Snap Connector

Step-by-Step Circuit Building Instructions:

- 1 Power Supply:** Connect the 9V battery snap connector to the breadboard. Connect the Red wire (+) to the Positive rail and Black wire (-) to the Ground rail.
- 2 Place the Potentiometer:** Insert the 3 pins of the 10kΩ Potentiometer into three separate rows in the middle section of the breadboard (e.g., rows 10, 12, and 14).
- 3 Connect Power Terminals:** Connect a jumper wire from the Positive rail (+) to Pin 1 (outer left pin) of the potentiometer. Connect Pin 3 (outer right pin) to the Ground rail (-). This creates a voltage divider across the dial!
- 4 Connect Output Signal:** Connect a jumper wire from Pin 2 (middle wiper pin) to a new column on the breadboard. This middle pin delivers the adjustable divided voltage.
- 5 Add Safety Resistor & LED:** Connect a 330Ω resistor from the middle pin jumper row to the long leg (+) of the LED. Connect the short leg (-) of the LED back to the Ground rail.
- 6 Test the Dimmer:** Slowly turn the potentiometer knob! Watch how controlling the voltage divides power and smoothly changes the LED brightness from dim to bright!

SAFETY FIRST: Ask an adult for help! Use small batteries only—NEVER connect circuit wires to wall outlets!

STUDENT PRACTICE WORKSHEET: VOLTAGE DIVIDER & POTENTIOMETER

Name: _____

Date: _____

Score: _____ / 6

Test Your Knowledge! (6 Questions)

1. Water Pipe Analogy: What does electricity flow like in our water pipe analogy, and what acts as the power source?

2. Concept Definition: What is a potentiometer, and how does turning its knob change the light bulb?

3. Fill in the Blanks:

a) Turning the potentiometer knob clockwise makes the light _____.

b) Turning the knob counter-clockwise makes the light _____.

c) Two or more resistors connected one after another are connected in _____.

4. Experiment Observation: When building your voltage divider circuit, which pin on the potentiometer provided the adjustable divided voltage to the LED?

5. Real-Life Application: Name one household item or appliance that uses a dial knob like a potentiometer (e.g., ceiling fan regulator).

6. Safety Rules: Why is it dangerous to plug classroom science projects directly into a wall power outlet?
